Ayam Banjade

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Education

Constructor University (formerly Jacobs University Bremen) **Sept. 2021-Present**
B.Sc. Computer Science, Mathematics minor — Bremen, Germany

- Notable Courses: Software Engineering, Database and Web Services, Algorithms and Data Structures, Operating Systems, Distributed Algorithms, Machine Learning, Automata Computability and Complexity, Artificial Intelligence, Programming in C/C++, Stochastic Methods Lab, Secure and Dependable Systems

Experience

Constructor University **Bremen, Germany**
Teaching Assistant - Numerical Methods — Feb. 2023 - Present

- Collaborated with other teaching assistants and the instructor to create and solve challenging problems for students, and give tutorials on:
  - root finding, interpolation, approximation
  - numerical differentiation, numerical integration (quadrature), and numerical solutions for differential equations

Teaching Assistant - Programming in C/C++ — Sept. 2022 - Dec. 2022

- Debugged and validated code, and gave tutorials in collaboration with other teaching assistants regarding:
  - pointers, pointer arithmetic, object-oriented programming, dynamic memory allocation, macros

Teaching Assistant - Introduction to Computer Science — Sept. 2022 - Dec. 2022

- Collaborated with the instructor and other teaching assistants to reduce students' doubt on:
  - functional programming (Haskell), assembly programming
  - discrete mathematics, boolean algebra, Hoare's logic
  - combinational and sequential digital circuits

Projects

Rippit
- Developed frontend and backend of a specialized forum website utilizing jQuery, and MySQL in MongoDB and PHP in collaboration with a team using Version Control System (git)
- Implemented using PHP to connect the SQL server and website using mysqli queries
- Increased functionality (suggestions, maps) using jQuery and the Leaflet library

Bug World
- Collaborated on various sprints to build a full stack simulation game written in JavaScript, HTML, CSS
- Developed a parser, assembler, simulator, and GUI using OOP, and implemented test cases using the Jest testing framework

Scrpr
- Interactive Discord bot that scrapes images and metadata from a static website developed in Python using `bs4`, and `requests` libraries and RegEx

Stochastic Methods Lab
- Developed stochastic models related to financial mathematics using Python's `matplotlib`, `numpy`, `pandas`, and `scipy` libraries and CSV manipulation

Spam Email Extracter
- Developed a program in C++ to extract 57 attributes based on UCI's Spambase Data Set to help detect spam emails

Skills

Programming Languages: C, C++, HTML, PHP, CSS, Javascript, Python, Rust, React, Bash, Lua, Shell Script, Kotlin, Java
Software: Linux and Linux Terminal, Windows, macOS, vim, Emacs, VSCode, MS Office Suite, Google Suite, LaTeX
Languages: English (fluent), Nepali (native), Hindi (conversational), German (A2)

Bremen, Germany Ayam